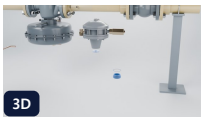
Fisher™ 1098-EGR regulator with 61LD pilot — Step-by-step

One page to follow along with the ② disassembly & assembly guide video — the ▶ time on the right opens that part of the video.

② Disassembly guide
English

Work by qualified personnel following the manufacturer's manual · KFNGS training material — not an Emerson publication
Work the manual does not describe: do not proceed until the manufacturer's procedure is confirmed
torques from the manual only · tool sizes from KFNGS service records — verify the actual fastener

Photo	Step · what to do / see	Tools · torque	★ Caution · check	Video
 3D	**Before you start — shutdown and pressure release** Slowly close the downstream block valve (and the control-line hand valve if its tap is downstream of it), then the upstream block valve and the pilot supply hand valve. Slowly vent the downstream piping and, where applicable, the control line completely; then slowly vent the upstream piping and pilot completely (p. 13).	—	Vent gas to a safe place (p. 8) · confirm zero on the gauges.	▶ 0:10
 3D	**① Remove the travel indicator assembly** Remove the protector (19) and scale (18), unscrew the lower indicator fitting (5) and lift out the assembly.	Fitting: 375 mm motor wrench **Assembly torque: 54 N·m (p. 15)**	For indicator-only work, leave the body flange in place (p. 15).	▶ 1:12
 Real 2006	**② Disassemble the indicator** Remove the nuts (22, 8) and separate the stem, spring, O-rings and backup rings. Leave the press-fit bushing (6) in place.	Nuts: 3/8 in. wrench	★ Spring compression remains and the fitting can jump — loosen the nuts slowly (p. 15).	▶ 1:26
 3D	**③ Body flange · trim package** Loosen the 8 cap screws (3) in a crisscross pattern and lift out the body flange with the trim. Inspect the gasket (4).	Cap screws: 3/4 in. wrench (e.g. NPS 2) **Assembly torque: NPS 2 = 75–95 N·m (Table 8 by size)**	Trim package = body flange, cage, seat ring and plug (Fig. 8).	▶ 1:56
 Real 2006	**④ Invert the body flange → cage** Invert the body flange and anchor it on the body (Fig. 10), then unscrew the cage (11). Remove the upper seal (15).	Strap wrench, or a soft bar through the windows	Inspect the cage O-ring (17) and replace it if needed (p. 14).	▶ 2:10
 3D	**⑤ Valve plug · piston ring · seat ring** Pull the plug (16), remove the piston ring (14) and O-ring (20), then unscrew the seat ring (13) from the cage. Remove the port seal (12).	A bar in the seat-ring slots (e.g. wrench handle)	★ Unfold the piston ring at its precut fold — take care not to break it.	▶ 2:28
 3D	**⑥⑦ 61LD pilot · control spring** Disconnect the tubing and unscrew the pilot from the nipple. Release spring-case pressure, open the closing cap and turn the adjusting screw counterclockwise to release compression.	Tubing nuts 18 mm · closing cap: motor wrench	★ Trapped pressure can make the cap fly off — release it first (61 Series p. 4). To change only the spring, leave the pilot in place.	▶ 2:50
 3D	**⑧–⑪ Pilot sections · diaphragms · yoke · 1806** Remove the 8 cap screws (P20) crisscross to split the pilot into three; inspect both diaphragms, yoke and relay orifice. Check that the 1806 spring closes the ball.	Cap screws: 1/2 in. wrench or socket **Assembly torque: 14–16 N·m, crisscross (p. 18)**	★ Reassemble the diaphragms without wrinkles — if the case rocks, suspect a wrinkle (p. 18).	▶ 3:23
 Real 2006	**⑫ Lower case · diaphragm · stem** Confirm the tubing is disconnected. Remove the cap screws (A10) and nuts (A11), detach the lower case, then remove the diaphragm, plate and stem assembly.	Cap screws: 14 mm **Torque: check the manual value**	Inspect the diaphragm (A7) and replace it if needed (kit).	▶ 4:25
 Real 2006	**⑬⑭ Nipples · tee · bonnet (stem O-ring replacement only)** Only when replacing stem O-rings: disconnect the control line, tee and nipples, then unscrew the bonnet (A3) with the upper case.	Grease fitting and vent 17 mm · bonnet: motor wrench **Torque: check the manual value**	Inspect the stem O-rings, wiper and bearings; replace if needed.	▶ 4:43
 3D	**Assembly · indicator zeroing · start-up** Assemble in reverse order. Fit the indicator fitting and set its zero, reconnect the tubing, then start up.	**Indicator fitting: 54 N·m (p. 15)**	★ Port and upper seals: 2–3 drops of silicone oil, flat side down (p. 16) · check the pipe plug (31) (p. 16).	▶ 5:31

Photos: Real = KFNGS service records (year · 299HS = parts shared with the 299H) · 3D = KFNGS guide screen (training reconstruction) · ①… = guide video steps

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Installation, operation and maintenance must be performed by qualified personnel following applicable codes, site safety procedures and the manufacturer's manual.
The disassembly & assembly guides are training reconstructions, not manufacturer procedures. Do not perform any work the manual does not describe, including releasing spring force, until you have checked the manufacturer's procedure.
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